

Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

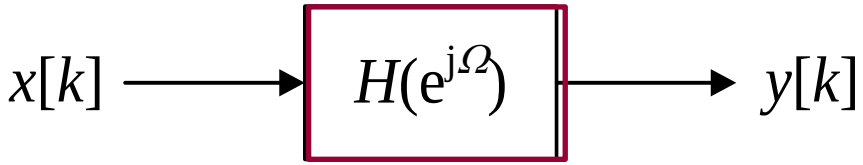
Zhou Hang

Introduction

- Digital filter concept and classification
- Technical requirements for digital filters
- The design goal of digital filter
- IIR digital filter design steps

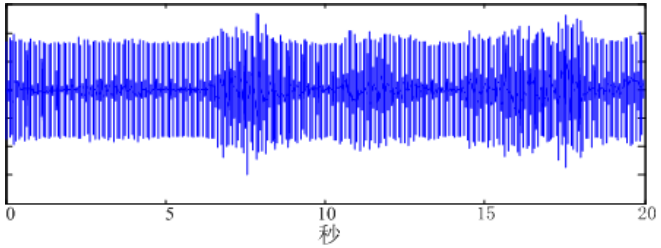
Digital filter concept and classification

Digital filter: It is a discrete-time system, which can selectively filter out some frequency components of the input signal according to the need, so as to realize the processing of the input signal.

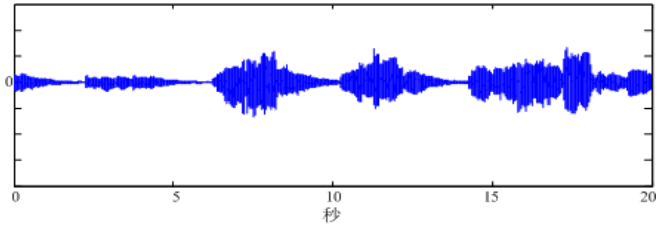


(Digital Filter, DF)

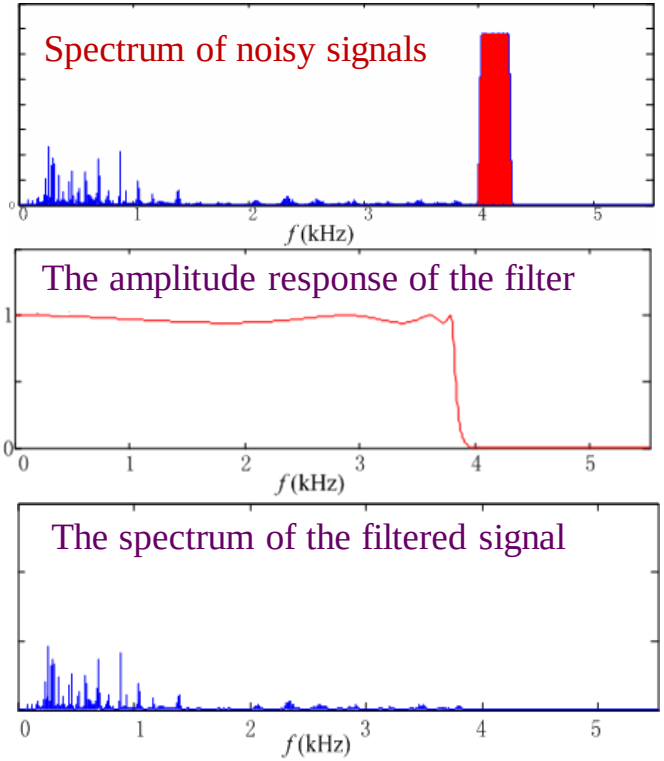
Digital filter concept and classification



Time domain waveform with noise signal



Time domain waveform of the filtered signal



Digital filter concept and classification

Functional division

- Low-pass filter
- High-pass filter
- Bandpass filter
- Band-stop filter

Divided by unit impulse response length

FIR filter

IIR filter

Digital filter concept and classification

LTI system $H(z)=\dfrac{\sum_{j=0}^{M} b_j z^{-j}}{1+\sum_{i=1}^{N} a_i z^{-i}}$

$a_1,\ldots,a_N$ At least one of the terms is non-zero,

$$H(z)=K\frac{(z-z_1)(z-z_2)\cdots(z-z_M)}{(z-p_1)(z-p_2)\cdots(z-p_N)}=\sum_{i=1}^{n}\frac{A_i z}{z-p_i}$$

$$h[k]=\mathcal{Z}^{-1}\{H(z)\}=\sum_{i=1}^{N}A_i(p_i)^k u[k]$$

Infinite in length

The system is called **IIR** (Infinite Impulse Response) system

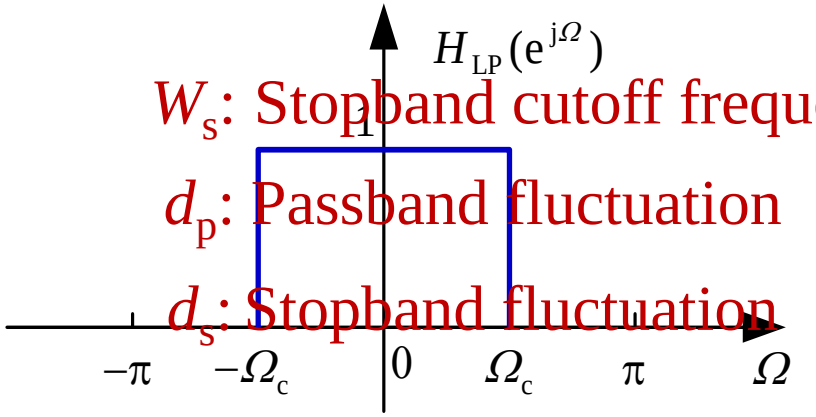
Digital filter concept and classification

W_p: Passband cut-off frequency

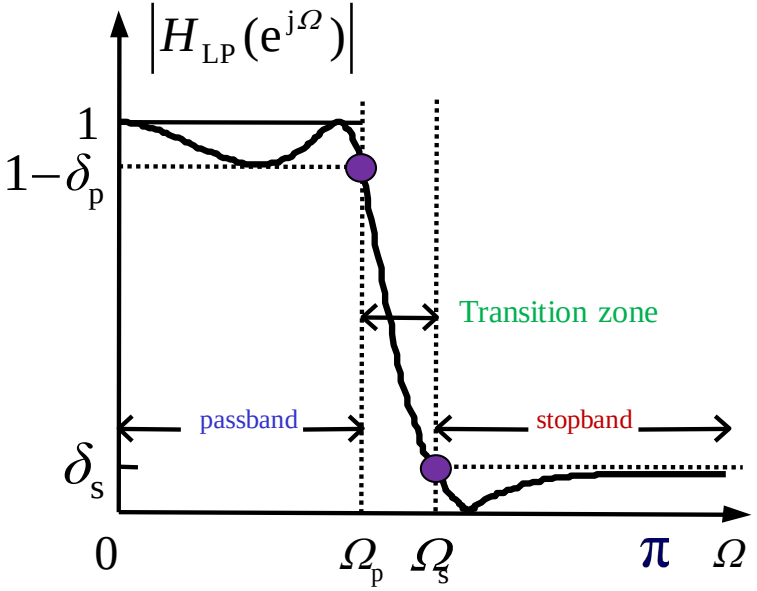
W_s: Stopband cutoff frequency

d_p: Passband fluctuation

d_s: Stopband fluctuation



Ideal low-pass digital filter



Actual low-pass digital filter

Maximum passband attenuation(dB) $A_p = -20\lg(1-\delta_p)$

Stopband minimum attenuation(dB) $A_s = -20\lg\delta_s$

Design goals for digital filters

LTI system $H(z)=\dfrac{\sum_{j=0}^{M} b_j z^{-j}}{1+\sum_{i=1}^{N} a_i z^{-i}}$

The system function $H(z)$ of the digital filter is obtained by determining M and N and the coefficients ai and bj from the given index of the digital filter frequency characteristics (Ω_p, A_p) and (Ω_s, A_s).

Such as $H(z)=\dfrac{1+2z^{-1}-3z^{-3}}{1-0.5z^{-1}}$

$b_0=1,\ b_1=2,\ b_2=0,\ b_3=-3;\quad M=3$

$a_1=-0.5;\quad N=1$

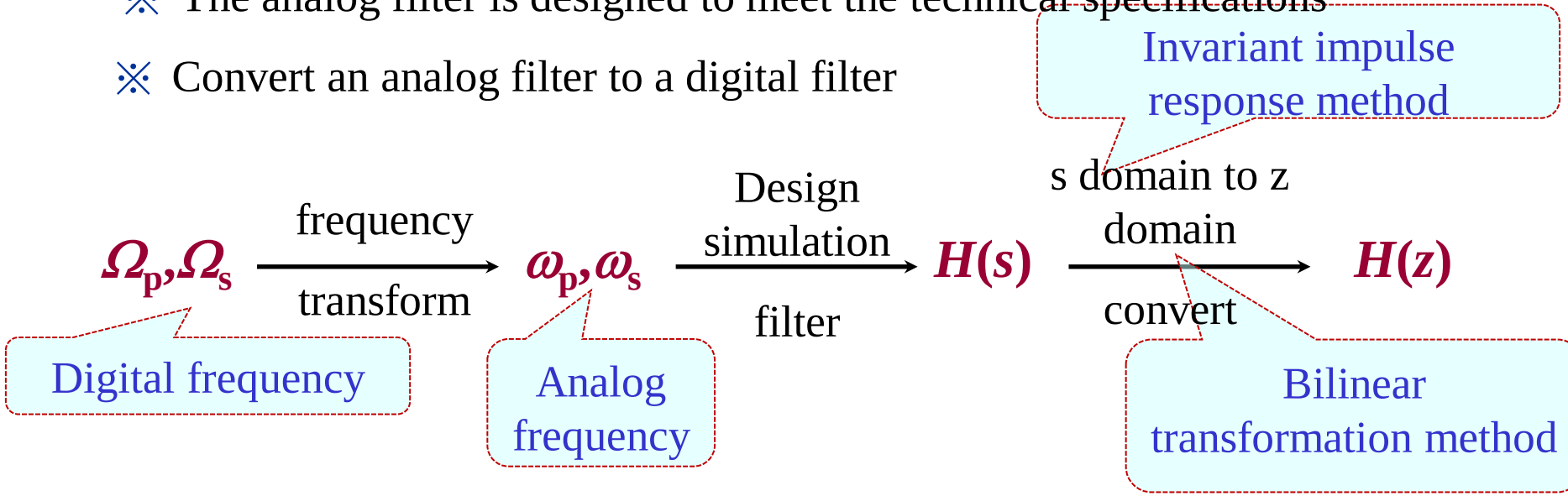
IIR digital filter design steps

The basic idea is to convert the design of digital filter into the design of analog filter

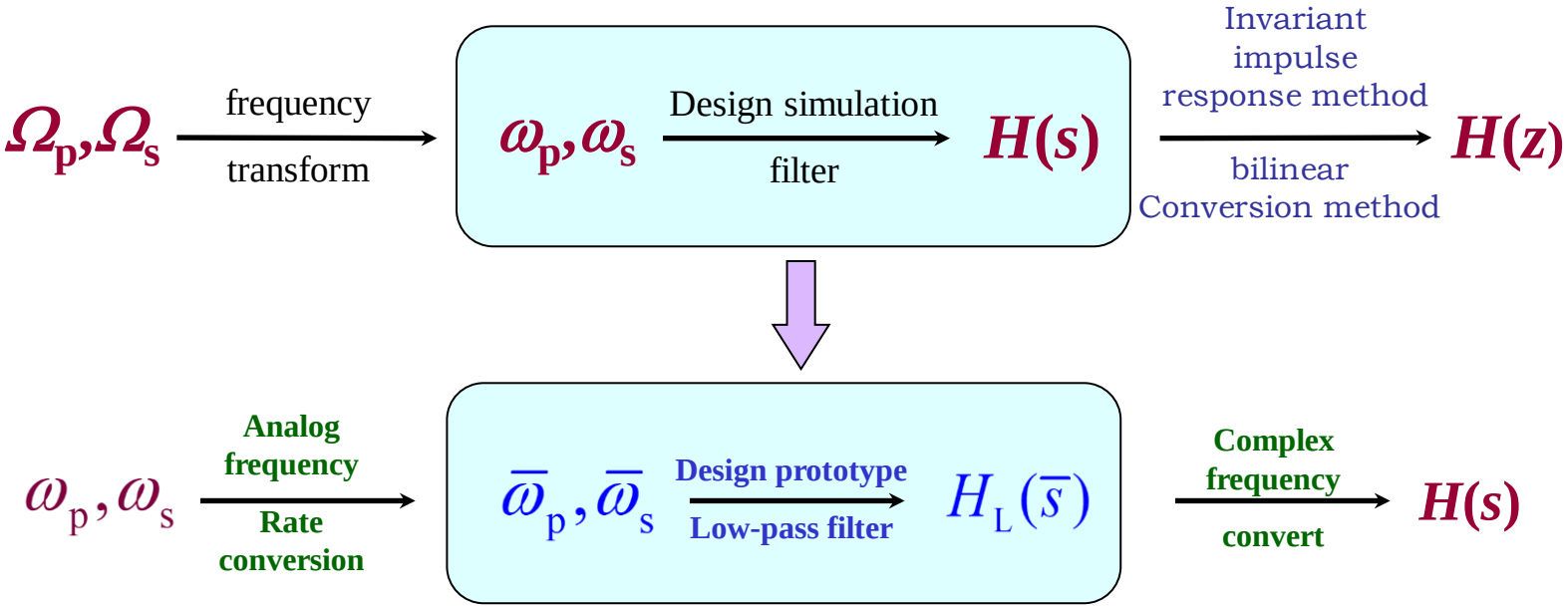
※ The index of the digital filter is converted to the index of the analog filter

※ The analog filter is designed to meet the technical specifications

※ Convert an analog filter to a digital filter



IIR digital filter design steps



BW、CB、C Three low-pass filter templates

Thank you

Some of the materials quoted in this course come from many years of teaching accumulation of the lecturer, from various media and the communication of colleagues and peers, it is difficult to specify the sources.